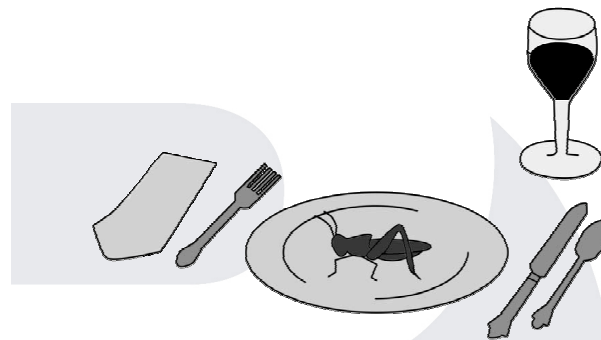


FOOD FOR THOUGHT

A *Why not eat insects?* So asked British entomologist Vincent M. Holt in the title of his 1885 treatise on the benefits of what he named *entomophagy* – the consumption of insects (and similar creatures) as a food source. The prospect of eating dishes such as “wireworm sauce” and “slug soup” failed to garner favour amongst those in the stuffy, proper, Victorian social milieu of his time, however, and Holt’s visionary ideas were considered at best eccentric, at worst an offense to every refined palate. Anticipating such a reaction, Holt acknowledged the difficulty in unseating deep-rooted prejudices against insect cuisine, but quietly asserted his confidence that “we shall some day quite gladly cook and eat them”.



B It has taken nearly 150 years but an eclectic Western-driven movement has finally mounted around the entomophagic cause. In Los Angeles and other cosmopolitan Western cities, insects have been caught up in the endless pursuit of novel and authentic delicacies. “Eating grasshoppers is a thing you do here”, bug-supplier Bricia Lopez has explained. “There’s more of a ‘cool’ factor involved.” Meanwhile, the Food and Agricultural Organization has considered a policy paper on the subject, initiated farming projects in Laos, and set down plans for a world congress on insect farming in 2013.

C Eating insects is not a new phenomenon. In fact, insects and other such creatures are already eaten in 80 per cent of the world’s countries, prepared in customary dishes ranging from deep-fried tarantula in Cambodia to bowls of baby bees in China. With the specialist knowledge that Western companies and organisations can bring to the table, however, these hand-prepared delicacies have the potential to be produced on a scale large enough to lower costs and open up mass markets. A new American company, for example, is attempting to develop pressurisation machines that would de-shell insects and make them available in the form of cutlets. According to the entrepreneur behind the company, Matthew Krisiloff, this will be the key to pleasing the uninitiated palate.

D Insects certainly possess some key advantages over traditional Western meat sources. According to research findings from Professor Arnold van Huis, a Dutch entomologist, breeding insects results in far fewer noxious by-products. Insects produce less ammonia than pig and poultry farming, ten times less methane than livestock, and 300 times less nitrous oxide. Huis also notes that insects – being cold-blooded creatures – can convert food to protein at a rate far superior to that of cows, since the latter exhaust much of their energy just keeping themselves warm.

E Although insects are sometimes perceived by Westerners as unhygienic or disease-ridden, they are a reliable option in light of recent global epidemics (as Holt pointed out many years ago, insects are “decidedly more particular in their feeding than ourselves”). Because bugs are genetically distant from humans, species-hopping diseases such as swine flu or mad cow disease are much less likely to start or spread amongst grasshoppers or slugs than in poultry and cattle. Furthermore, the squalid, cramped quarters that encourage diseases to propagate among many animal populations are actually the residence of choice for insects, which thrive in such conditions.

F Then, of course, there are the commercial gains. As FAO Forestry Manager Patrick Durst notes, in developing countries many rural people and traditional forest dwellers have remarkable knowledge about managing insect populations to produce food. Until now, they have only used this knowledge to meet their own subsistence needs, but Durst believes that, with the adoption of modern technology and improved promotional methods, opportunities to expand the market to new consumers will flourish. This could provide a crucial step into the global economic arena for those primarily rural, impoverished populations who have been excluded from the rise of manufacturing and large-scale agriculture.

G Nevertheless, much stands in the way of the entomophagic movement. One problem is the damage that has been caused, and continues to be caused, by Western organisations prepared to kill off grasshoppers and locusts – complete food proteins – in favour of preserving the incomplete protein crops of millet, wheat, barley and maize. Entomologist Florence Dunkel has described the consequences of such interventions. While examining children’s diets as a part of her field work in Mali, Dunkel discovered that a protein deficiency syndrome called kwashiorkor was increasing in incidence. Children in the area were once protected against kwashiorkor by a diet high in grasshoppers, but these had become unsafe to eat after pesticide use in the area increased.

H A further issue is the persistent fear many Westerners still have about eating insects. “The problem is the *ick* factor—the eyes, the wings, the legs,” Krisiloff has said. “It’s not as simple as hiding it in a bug nugget. People won’t accept it beyond the novelty. When you think of a chicken, you think of a chicken breast, not the eyes, wings, and beak.” For Marcel Dicke, the key lies in camouflaging the fact that people are eating insects at all. Insect flour is one of his propositions, as is changing the language of insect cuisine. “If you say it’s mealworms, it makes people think of ringworm”, he notes. “So stop saying ‘worm’. If we use Latin names, say it’s a *Tenebrio* quiche, it sounds much more fancy”. For Krisiloff, Dicke and others, keeping quiet about the gritty reality of our food is often the best approach.

I It is yet to be seen if history will truly redeem Vincent Holt and his suggestion that British families should gather around their dining tables for a breakfast of “moths on toast”. It is clear, however, that entomophagy, far from being a kooky sideshow to the real business of food production, has much to offer in meeting the challenges that global societies in the 21st century will face.